

TRACCO BUS MANAGEMENT

DATABASE DESIGN & SCHEMA REPORT

PostgreSQL Database Schema Specifications

A comprehensive guide to the tables, columns, constraints, and relationships.

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Date: August 2026

Version: 1.0.0 (Production Schema)

1. Executive Summary & Architecture

The Tracco Bus Management database is designed using PostgreSQL to support complex logistics, student subscription billing, driver allocations, and maintenance history tracking. The database consists of 25 tables which are divided into distinct functional areas:

- Transport Logistics: Management of routes, stop points, and route mappings.
- Student Subscriptions: Academic profiles and fee histories (B.Tech and M.Tech subscriptions).
- Fleet & Operations: Bus metadata, GPS/odometer readings, and document compliances.
- Maintenance & Consumables: Refueling (diesel), oil replacement logs, and spare part replacements.
- Personnel & Security: Driver registration, driver contacts, and administrator user credentials.

Key Database Relationships

- Routes to Buses: 1-to-many relationship mapping which bus runs on which route.
- Routes to Stoppings: M-to-N relationship mapped through the route_stop_map bridge table.
- Students to Boarding: Students link to route_stop_map representing their specific boarding stop.
- Students to Fee History: Fee history log tracking payments made per semester.
- Buses to Logs: 1-to-many relationship tracking fuel (diesel_logs), oil (oil_logs), and spares (spare_usage).
 - Spare Parts Management: spare_stocks maps to spare_purchases and individual spare_items. spare_usage_details links specific items to usage incidents.

2. Table-by-Table Schema Specifications

2.1 Logistics & Routing Subsystem

2.1.1 `routes`

2.1.2 `stoppings`

2.1.3 `route\_stop\_map`

2.2 Student & Transport Subscriptions Subsystem

2.2.1 `branch`

2.2.2 `btech\_students`

2.2.3 `btech\_students\_bus\_fee\_history`

2.2.4 `mtech\_students`

2.2.5 `mtech\_students\_bus\_fee\_history`

2.3 Fleet & Operations Subsystem

2.3.1 `buses`

2.3.2 `bus\_readings`

2.3.3 `bus\_documents`

2.3.4 `document\_types`

2.4 Consumables & Maintenance Log Subsystem

2.4.1 `fuel\_rates`

2.4.2 `diesel\_logs`

2.4.3 `oil\_stocks`

2.4.4 `oil\_logs`

2.5 Inventory & Spare Parts Subsystem

2.5.1 `spare\_stocks`

2.5.2 `spare\_purchases`

2.5.3 `spare\_items`

2.5.4 `spare\_usage`

2.5.5 `spare\_usage\_details`

2.6 Personnel & Authentication Subsystem

2.6.1 `drivers`

2.6.2 `drivers\_phone\_number`

2.6.3 `users`

2.6.4 `test\_inventory`